

FHA Insurance Premiums

Cello Research

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1 Introduction

The **FHA Insurance premium** is a **mortgage insurance premium (MIP)** that is used to cushion the FHA against losses incurred over the course of mortgage foreclosures. The FHA insurance premium paid by borrowers consists of two separate amounts: (1) a lump sum at closing (the **upfront mortgage insurance premium** or **UFMIP**), and (2) a monthly fee that can extend over several years (the **annual premium**). Both sums are charged as a percentage of the initial loan amount and will be discussed in further detail below.

Any MIP adds to the cost of the mortgage and monitoring changes in these costs is important from the perspective of forecasting prepayments since these changes reflect fundamental shifts in the borrower's hurdle for refinancing. As a matter of fact, the FHA UFMIP and the annual premium have all been subject to several revisions over the years.

Our focus here is three-fold: (1) A detailed description of the components of FHA insurance premiums; (2) A catalog of the changes in these insurance premiums (for a streamline refinancing) over the years; and (3) Specifying how changes in the FHA insurance premium could be incorporated into a prepayment model. The baseline mortgage contract we will focus on represents the majority of all outstanding FHA mortgages: a 30-year mortgage with a current (amortized) loan-to-value (LTV) ratio of greater than 95%. Furthermore, we will only discuss insurance premiums for streamline refinancings (SRs) since HUD data show that 80%-90% of all FHA-to-FHA refinancings are SRs.

2 UFMIP

The upfront premium is intended to be paid at the origination of the mortgage and is expressed as a fraction of the loan amount. In practice, the FHA allows borrowers to finance the premium by rolling it into their loan balance, in essence allowing the upfront premium to be paid on the same schedule as the principal balance. Almost all borrowers finance their upfront premiums, although a cash-out option is also available. When added to the mortgage, the UFMIP amount is not taken into account when determining compliance with statutory loan limits or LTV restrictions – only the base mortgage amount must comply with these requirements.

The upfront premium can be **refunded** based on certain criteria being satisfied. FHA first introduced the upfront premium refund program in 1983. Over time, the FHA “earns” some portion of the UFMIP and the unearned portion is refunded to the borrower based on the elapsed time period between the origination and refinancing dates. When the refund program was first introduced, borrowers could receive a refund whenever they prepaid their mortgages prior to the maturity of the mortgage contract. In 2000, FHA changed its policy so that a borrower would only obtain a refund if they prepaid within the first five years of their mortgage contract. As of 2004, only if the borrower is refinancing their current FHA loan to another FHA loan within

3 years following the original endorsement date will a refund credit be applied to reduce the amount of the UFMIP paid on the new loan.¹

The historical evolution of the UFMIP and the Annual MIP over time is depicted in Table 1.

3 Annual MIP

The amount of the annual premium is based upon the *amortized* unpaid principal balance (excluding any upfront premiums) and is collected on a monthly basis.² For SRs, the charge is currently set based on the LTV and the term of the mortgage.³ In particular, the MIP is cancelled when the LTV reaches 78% for 30-year mortgages, provided that the borrower has paid the annual MIP for at least 5 years.⁴

Table 1 shows the historical evolution of the UFMIP and the Annual MIP over time.

4 Modeling Premium Changes

Table 1 in Appendix A shows that the insurance premiums paid by FHA borrowers who are pursuing a streamline refinancing are a complicated function of:⁵

1. **Calendar time:** Insurance premiums have both increased and decreased over time.
2. **Origination date of the loan:** At any given moment in time, borrowers that wish to refinance can be charged differing insurance premiums depending upon when their loan was endorsed.⁶
3. **Loan Size:** As of April 2012, loans with balances greater than \$625,500 face higher annual premiums.
4. **LTV:** Loans with higher LTVs pay higher annual premiums.
5. **Seasoning:** Only loans with less than three years of seasoning receive a refund of the UFMIP that can be applied towards the upfront charge on the new loan.

We can categorize these premium changes as follows: (1) Changes in annual premiums; (2) Changes in UFMIPs; and (3) Changes in the UFMIP refund schedule; and articulate a modeling approach for each such category.

4.1 Modeling Changes in Annual Premiums

Even though the annual MIP is expressed as an interest rate charge, it does not appear directly in an FHA mortgage rate quote and is an add-on to this quote. In any case, the core point is that a change in the annual premium directly translates into a change in the overall interest rate available to a potential refinancer. A

¹To see the refund schedule for the three year case, see p. 6 of http://portal.hud.gov/hudportal/documents/huddoc?id=4155-2_7.pdf. Approximately 58% of the UFMIP is refunded after 12 months and 34% after 24 months.

²Note that the annual premium payment decreases over time as the principal of the mortgage amortizes.

³The insurance premium also depends upon the loan purpose and the loan type (ARM or FRM).

⁴Note that since the LTV ratio is *not* based on the appreciation of the property, it will typically take about 10-12 years for most loans to reach this target level since the average starting LTV on an FHA loan is 95% or more.

⁵The dependence of premiums on loan type and term is not relevant from the perspective of modeling refinancings of FHA-to-FHA 30-year mortgages. It may however be relevant in modeling turnover.

⁶The endorsement date is not the same as the closing date; it corresponds to the date that FHA endorsed the borrower for insurance coverage. Loan documents are sent to the FHA within 60 days of the closing date but in some cases loans can be endorsed up to a year after they are closed. In the typical situation, a loan is endorsed within two months of closing. However, for our purposes, we will assume that loan origination coincides with endorsement.

50bps annual MIP is not the same as adding 50bps to the mortgage rate but we can make the heuristic assumption that they can be treated as equal.⁷ Since the MIP is not captured in the interest rate, statutory changes in the MIP rate over time will need to be exogenously specified in a prepayment model via the incentive function. Table ?? shows the effective change in incentive (a positive number implies that the borrower needs a greater incentive to refinance) for loans corresponding to different origination dates. Greater realism can be achieved by also modeling the fact that changes in the annual MIP also depend on LTV and loan size.

Table 1: **Annual MIP-related Changes in Incentive by Origination Date**

Orig. Date	Ann. MIP	Oct 2008	Oct 2010	Apr 2011	Apr 2012	Jun 2012
pre-Oct 2008	50bps	+5bps	+40bps	+65bps	+75bps	+5bps
Oct 2008 - Jun 2009	55		+35	+60	+70	+5
Apr 2010 - Oct 2010	55		+35	+60	+70	+70
Nov 2010 - Apr 2011	90			+25	+35	+35
May 2011 - Apr 2012	115				+10	+10
May 2012 -	125					+0

Source: Cello calculations based on FHA data

4.2 Modeling Changes in UFMIPs

The UFMIP is rolled into the mortgage principal and therefore affects the monthly payment through that channel.⁸ Understanding how much this increase in the balance equates to in mortgage rate space depends upon the borrower's tenure horizon. For example, on the typical 30-year loan, 1% of the principal balance is approximately equal to a 25bps change in rate and assumes that the typical borrower has a tenure horizon of 4-5 years. If it not completely clear if this is the appropriate framework for looking at the UFMIP cost because, unlike the case with rate-point conversions, the UFMIP amount can be rolled into the mortgage balance. From the lens of a change in the monthly mortgage payment, a 1% increase in the balance is effectively equivalent to an 8bps increase in the mortgage rate. In either case, we have to model changes in the UFMIP as time-varying changes in transaction costs for a refinancing borrower. These costs do depend on the origination year because of grandfathering effects (as can be see in Table 1).

4.3 Modeling Changes in the UFMIP Refund Schedule

The refund schedule has not changed since 2004. It can be captured as a scaling factor applied to the UFMIP as a function of loan age (and time, if necessary). However, this effect is not easy to capture directly because, given a GNMA pool, we do not know the distribution of the initial UFMIP's that various borrowers have been charged, some fraction of which would potentially be refunded in the case of a refinancing. One can make the reasonable assumption that any refinancing patterns that are age-dependent should show up in calibrating the refinancing seasoning ramp for the model.

Appendix A: Premium Rates for Streamline Refinance Loans

We document changes in premium rates for FHA streamline refinancings. As discussed in the Introduction, the focus is on 30-year mortgages with LTVs greater than 95%. For a given row int he table, all FHA loans

⁷It is not the same since the monthly amount of the MIP is added to the standard monthly P&I payment without trying to keep the payment level. Also, the annual premium is automatically cancelled once the amortized principal balance reaches 78% of the original appraisal, provided the borrower has paid the annual MIP for at least 5 years.

⁸As we know, the amount of the monthly payment on a level-payment fixed-rate mortgage is a function of the interest rate, principal, loan amount, and loan term.

originated in the Calendar Date range specified in the first column of the table which then subsequently execute a streamline refinance pay the premiums specified in the adjacent columns. Also, maintaining our focus on the same row, FHA loans originated prior to this date but that execute a SR within this date range may or may not pay the new premiums specified in the adjacent columns depending upon whether their existing insurance premiums are grandfathered or not. Also, the insurance premiums upon refinancing for this pre-existing collection of loans can sometimes be retrospectively revised. While these types of events happen rarely they can have a significant impact. Here are two examples:

- **January 1, 2001 change.** Only applies to mortgages originated after this date. SRs of mortgages closed before July 1991 have their annual premiums grandfathered: i.e., they remain exempt from the annual premium and will be charged an upfront premium of only 1.50%.⁹
- **April 2012 change.** The annual premium increases from 115bps to 125bps for most loans. However, loans originated prior to May 2009 and after July 2008 have their annual premiums grandfathered at 55bps. The UFMIP increases from 1% to 1.75% regardless of loan size and LTV for most loans with the exception of pre-May 2009 loans, where the UFMIP will decreased from 1% to 0.01%. This specific change applies from June 2012.

Table 2: The Evolution of FHA Insurance Premiums for Streamline Refinancings

Endorsement Date	UFMIP	Annual MIP
Prior to 9/1/1983	None	None
9/1/83 to 6/30/91	3.80%	None
7/1/91 to 9/30/92	3.80%	0.5% for first 7 years
10/1/92 to 4/16/94	3.00%	0.5% for first 7 years
4/17/94 to 12/31/00	2.25%	0.5% for first 7 years
1/1/01 to 09/30/08	1.50%	0.5% until loan balance reaches 78% amortized LTV, minimum of 5 years
10/1/08 to 03/31/10	1.50%	0.55% for LTV > 95%; 0.50% otherwise until loan reaches 78% LTV
04/1/10 to 10/03/10	2.25%	0.55% for LTV > 95%; 0.50% otherwise until loan reaches 78% LTV
10/4/10 to 4/17/11	1%	0.90% for LTV > 95%; 0.85% otherwise until loan reaches 78% LTV
4/18/11 to 4/8/12	1%	1.15% for LTV > 95%; 1.10% otherwise until loan reaches 78% LTV
4/9/12 onwards	1.75% except for SRs of loans originated before June 1, 2009 which pay 0.01% starting June 11, 2012	1.25% for LTV > 95%; 1.20% otherwise. Add 25bps if loan size greater than \$625,500. For loans originated before June 1, 2009 Annual MIP is 0.55% starting June 11, 2012. MIP cancelled at 78% LTV.

Source: FHA

⁹To keep the table at a manageable size, this information is recorded therein.